A Closed Form for the Screw Rotation of a Segmented Helix

A companion note to the Joint Face Normal Method

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Abstract

The Joint Face Normal Method constructs the rigid motion between successive identical subunits of a periodic chain. Its rotational part admits a simple closed form. If α_j is the smallest rotation that opposes the two joint-face normals and τ_{eff} is the subsequent twist about the joint line, then

$$\boxed{\cos\frac{\theta}{2} = \cos\frac{\alpha_j}{2}\left|\cos\frac{\tau_{\text{eff}}}{2}\right|.}$$

Here $\theta \in [0, \pi]$ is the principal rotation angle of the full rigid motion. We prove the result, specify its twist convention, and give exact expressions reproducing all 23 nonzero-radius rows in the original Platonic helix catalog. Five straight-joint cases require distinguishing the rotation of the subunits from the geometry of the joint points. Translation and the remaining helix parameters are left for subsequent work.

1 Purpose and terminology

This note records a result for discussion with colleagues and for the segmented-helixes repository. It supplements, rather than revises, the 2022 paper [1]. The relevant construction is the *Joint Face Normal Method* in Sections 2.15–2.17 of that paper. We retain its joint points B, C, normalized face normals $\vec{N}_B, \vec{N}_C$, joint plane, joint line, joint twist τ, and screw rotation θ.

Define the *joint-face alignment angle*

$$\alpha_j = \arccos(-\vec{N}_B \cdot \vec{N}_C), \qquad 0 \le \alpha_j \le \pi. \tag{1}$$

This was called α in the exploratory discussion; the subscript avoids confusion with the original paper's $\alpha(\tau)$ in the Twist Spectrum proof. For adjacent faces of a convex polyhedron, α_j is its interior dihedral angle.

The essential convention is that τ_{eff} is measured *after the shortest normal-aligning rotation.* When an existing out-vector or software reference agrees with that choice, $\tau_{\text{eff}} = \tau$. A reference phase may otherwise be needed; Section 3 defines it without fitting it to a table.

2 The joint-face rotation theorem

Theorem 1 (Joint-face rotation)**.** *Suppose $0 < \alpha_j < \pi$. Let A be the shortest rotation sending $\vec{N}_B$ to $-\vec{N}_C$. After this alignment, apply a twist τ_{eff} about the joint line, using $\vec{N}_C$ as its oriented axis. For the resulting rotational part $R = \text{Rot}(\vec{N}_C, \tau_{\text{eff}})A$, its principal angle $\theta \in [0, \pi]$ satisfies*

$$\cos\frac{\theta}{2} = \cos\frac{\alpha_j}{2}\left|\cos\frac{\tau_{\text{eff}}}{2}\right|, \tag{2}$$

$$\cos\theta = \frac{(1+\cos\alpha_j)(1+\cos\tau_{\text{eff}})}{2} - 1. \tag{3}$$

For $-\pi \le \tau_{\text{eff}} \le \pi$, the absolute value in (2) can be omitted.

Proof. The shortest alignment has angle α_j and unit axis

$$\vec{u} = \frac{\vec{N}_B \times (-\vec{N}_C)}{\sin\alpha_j}.$$

Thus $\vec{u} \cdot \vec{N}_C = 0$: the alignment axis is perpendicular to the joint line. Choose a right-handed orthonormal basis in which $\vec{u}$ is the x-axis and $\vec{N}_C$ is the z-axis. This conjugates R and preserves its trace. In these coordinates,

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha_j & -\sin\alpha_j \\ 0 & \sin\alpha_j & \cos\alpha_j \end{pmatrix}, \qquad D = \begin{pmatrix} \cos\tau_{\text{eff}} & -\sin\tau_{\text{eff}} & 0 \\ \sin\tau_{\text{eff}} & \cos\tau_{\text{eff}} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplying $R = DA$ gives

$$\operatorname{tr} R = \cos\tau_{\text{eff}} + \cos\tau_{\text{eff}}\cos\alpha_j + \cos\alpha_j.$$

Since a proper three-dimensional rotation obeys $\cos\theta = (\operatorname{tr} R - 1)/2$, this proves (3). Factoring gives

$$1 + \cos\theta = \frac{(1+\cos\alpha_j)(1+\cos\tau_{\text{eff}})}{2}, \qquad \cos^2\frac{\theta}{2} = \cos^2\frac{\alpha_j}{2}\cos^2\frac{\tau_{\text{eff}}}{2}.$$

Taking the nonnegative square root proves (2). □

The homogeneous rigid motion is $T = \left(\begin{smallmatrix} R & \vec{t} \\ 0 & 1 \end{smallmatrix}\right)$. Its translation $\vec{t}$ therefore cannot change θ. Equivalently, the quaternion product has scalar part $\cos(\alpha_j/2)\cos(\tau_{\text{eff}}/2)$, because its constituent axes are perpendicular. The absolute value handles the two quaternion representatives of the same rotation.

Consequently $\theta(\tau_{\text{eff}}) = \theta(-\tau_{\text{eff}})$, $\theta(0) = \alpha_j$, and $\alpha_j \le \theta \le \pi$. These statements concern rotation magnitude; they do not determine handedness, axial travel, or whether the chain closes.

3 Relating the theorem to the paper's twist

3.1 A reference phase for any normal-aligning baseline

Let R_0 be the zero-twist rotation prescribed by an out-vector convention or implementation. It makes the faces flush, so $R_0\vec{N}_B = -\vec{N}_C$. The rotation $G = R_0A^{-1}$ fixes $\vec{N}_C$. Hence it is a rotation about that normal: $G = \mathrm{Rot}(\vec{N}_C, \gamma_j)$, where γ_j is defined modulo 2π. Applying the paper's additional twist gives

$$R = \mathrm{Rot}(\vec{N}_C, \tau)R_0 = \mathrm{Rot}(\vec{N}_C, \tau + \gamma_j)A, \qquad \boxed{\tau_{\mathrm{eff}} = \tau + \gamma_j \pmod{2\pi}.} \tag{4}$$

The theorem now applies. This phase depends on the chosen reference; it is not determined by the two normals alone. For any unit vector $\vec{v} \perp \vec{N}_C$, it can be measured by

$$\gamma_j = \mathrm{atan2}\Big(\vec{N}_C \cdot (\vec{v} \times G\vec{v}),\ \vec{v} \cdot G\vec{v}\Big).$$

Thus a discrepancy of twist conventions can be resolved from the baseline rotation or the face-attached out vectors, rather than inferred from a desired value of θ.

3.2 Why the Platonic checks use the tabulated twist directly

The repository's `adjoinPrism` uses two shortest rotations through the object-axis direction $\vec{e} = (C - B)/\|C - B\|$. Write $S(\vec{a}, \vec{b})$ for the shortest rotation from $\vec{a}$ to $\vec{b}$. Its forward baseline is

$$R_0 = S(\vec{e}, \vec{N}_C)\, S(\vec{N}_B, -\vec{e}). \tag{5}$$

For the face centers of a regular solid, with its center taken as the origin, $B = \rho\vec{N}_B$ and $C = \rho\vec{N}_C$ for the same $\rho > 0$. Therefore

$$\vec{e} = \frac{\vec{N}_C - \vec{N}_B}{\sqrt{2(1 - s)}}, \qquad s = \vec{N}_B \cdot \vec{N}_C.$$

Both rotations in (5) have axis parallel to $\vec{N}_B \times (-\vec{N}_C)$, and

$$\vec{N}_B \cdot (-\vec{e}) = \vec{e} \cdot \vec{N}_C = \sqrt{\frac{1 - s}{2}} = \cos\frac{\alpha_j}{2}.$$

Each has angle $\alpha_j/2$. Their product is exactly A, so $\gamma_j = 0$. This establishes the reference convention for the nonparallel Platonic checks from geometry and the implementation, independently of their numerical agreement.

For an arbitrary prism, the object axis need not lie in the plane of the two normals, and this two-stage baseline may have $\gamma_j \neq 0$. For example, with $\vec{e} = (0, 0, 1)$, $\vec{N}_B = (1, 0, -1)/\sqrt{2}$, and $\vec{N}_C = (0, 1, 1)/\sqrt{2}$, the alignment angle is 60°, but (5) has principal angle 62.7994° at $\tau = 0$. Its phase is $\gamma_j = \arccos(2\sqrt{2}/3) = 19.4712°$. The effective-twist formula gives the correct value.

4 Face classes without arbitrary face numbers

Fix the Base Joint Face B and classify distinct Chained Joint Faces C by the invariant $\vec{N}_B \cdot \vec{N}_C$. The classes below replace the software's arbitrary C-face numbering. A denotes adjacent, S, T, F successive normal-dot-product classes, and O opposite. Multiplicities count faces, not the catalog's shape analogs.

Solid	Class	Number of faces	$\cos\alpha_j = -\vec{N}_B \cdot \vec{N}_C$
Tetrahedron	A	3	$1/3$
Cube	A	4	0
	O	1	1
Octahedron	A	3	$-1/3$
	S	3	$1/3$
	O	1	1
Dodecahedron	A	5	$-1/\sqrt{5}$
	S	5	$1/\sqrt{5}$
	O	1	1
Icosahedron	A	3	$-\sqrt{5}/3$
	S	6	$-1/3$
	T	6	$1/3$
	F	3	$\sqrt{5}/3$
	O	1	1

4.1 The dodecahedron from its dual

Put $\varphi = (1+\sqrt{5})/2$. Dodecahedral face normals point toward the vertices of an icosahedron. Use the twelve vectors

$$(0, \pm 1, \pm\varphi), \quad (\pm 1, \pm\varphi, 0), \quad (\pm\varphi, 0, \pm 1),$$

all divided by $\sqrt{1+\varphi^2}$, and fix $\vec{N}_B = (0,1,\varphi)/\sqrt{1+\varphi^2}$. Their dot products with $\vec{N}_B$, excluding B itself, are $1/\sqrt{5}, -1/\sqrt{5}, -1$ with counts $5, 5, 1$. For example, $(1, \varphi, 0)$ yields $\varphi/(1+\varphi^2) = 1/\sqrt{5}$.

4.2 The icosahedron from its dual

Icosahedral normals point toward the twenty vertices of a dodecahedron. Take $(\pm 1, \pm 1, \pm 1)$ and

$$(0, \pm\varphi^{-1}, \pm\varphi), \quad (\pm\varphi^{-1}, \pm\varphi, 0), \quad (\pm\varphi, 0, \pm\varphi^{-1}),$$

all divided by $\sqrt{3}$. With $\vec{N}_B = (1,1,1)/\sqrt{3}$, the other nineteen dot products are

$$\sqrt{5}/3, \quad 1/3, \quad -1/3, \quad -\sqrt{5}/3, \quad -1,$$

with counts $3, 6, 6, 3, 1$. These facts follow directly from $\varphi + \varphi^{-1} = \sqrt{5}$ and $\varphi - \varphi^{-1} = 1$.

5 Exact catalog checks

All twists and catalog values below are from Table 1 of the original paper and the supplied LaTeX source [1, 2]. The convention established in Section 3 gives $\tau_{\text{eff}} = \tau$. Every entry means $\theta = \arccos(\text{listed exact expression})$; angles in the last two columns are in degrees. Numerical agreement is checked at the catalog's three-decimal precision. The proof, rather than agreement alone, establishes the identity.

5.1 Tetrahedral, cubic, and octahedral cases

Name	Class	τ	Exact $\cos\theta$	Computed	Catalog
Tetratorus	Tet A	0	$1/3$	70.5288	70.529
Tetrahelix	Tet A	-120	$-2/3$	131.8103	131.810
Cubatorus	Cub A	0	0	90.0000	90.000
Blockhelix	Cub A	-90	$-1/2$	120.0000	120.000
Octagear	Oct A	0	$-1/3$	109.4712	109.471
Octaspiky	Oct A	-120	$-5/6$	146.4427	146.443
Treestar	Oct S	0	$1/3$	70.5288	70.529
Octamedium	Oct S	-120	$-2/3$	131.8103	131.810

The octahedral names follow the supplied manuscript. The face classes and exact values remain unambiguous if labels differ elsewhere.

5.2 Dodecahedral cases

Name	Class	τ	Exact $\cos\theta$	Computed	Catalog
Dodecagear	A	0	$-1/\sqrt{5}$	116.5651	116.565
Dodecashaft	A	-72	$(\sqrt{5}-15)/20$	129.6575	129.657
Dodecadoubler	A	-144	$-1/2-1/\sqrt{5}$	161.3006	161.301
Dodecadonut	S	0	$1/\sqrt{5}$	63.4349	63.435
Dodecacorkscrew	S	-72	$-1/2+1/\sqrt{5}$	93.0258	93.026
The Alternater	S	-144	$-3/4-\sqrt{5}/20$	149.5197	149.520

For example, Dodecacorkscrew has $\cos\alpha_j = 1/\sqrt{5}$ and $\cos\tau = \cos 72^\circ = (\sqrt{5}-1)/4$. Therefore

$$\cos\theta = \frac{(1+1/\sqrt{5})(1+(\sqrt{5}-1)/4)}{2} - 1 = -\frac{1}{2} + \frac{1}{\sqrt{5}}.$$

Thus $\theta = \arccos(-1/2+1/\sqrt{5}) = 93.0258445^\circ$. There are six nonzero-radius dodecahedral checks; Dodecabeam is discussed in Section 6.

5.3 Icosahedral cases

Name	Class	τ	Exact $\cos\theta$	Computed	Catalog
Icosa Tree Star	A	0	$-\sqrt{5}/3$	138.1897	138.190
Two Strands	A	120	$-(9+\sqrt{5})/12$	159.4456	159.446
Planar point cluster	S	0	$-1/3$	109.4712	109.471
Rock Candy	S	120	$-5/6$	146.4427	146.443
Big Icosacorkscrew	T	45	$(\sqrt{2}-1)/3$	82.0638	82.064
Icosacorkscrew	T	-75	$(\sqrt{6}-\sqrt{2}-2)/6$	99.2526	99.253
Quasi-planar	T	165	$-(2+\sqrt{6}+\sqrt{2})/6$	167.7643	167.764
The Wheel	F	0	$\sqrt{5}/3$	41.8103	41.810
Slow Twist	F	120	$(\sqrt{5}-9)/12$	124.3094	124.309

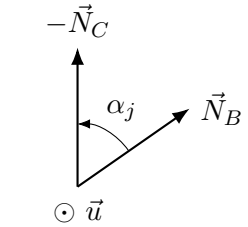
The three class-T twists $45°, -75°, 165°$ are separated by $120°$, as triangular vertex matching requires. They test the same face geometry with three distinct orientations. For Icosacorkscrew,

$$\cos\theta = \frac{(1+1/3)(1+\cos 75°)}{2} - 1 = \frac{\sqrt{6}-\sqrt{2}-2}{6},$$

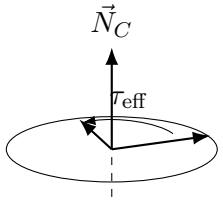
using $\cos 75° = (\sqrt{6}-\sqrt{2})/4$.

5.4 What is being verified

Together, these tables give $2+2+4+6+9 = 23$ checks with nonzero joint-helix radius. For such a joint orbit, the angular advance of consecutive joints is the principal rotation angle of R. No reference phase was chosen to force these matches: the normals supply α_j, the original catalog supplies τ, and the face-center argument supplies $\gamma_j = 0$.



Shortest alignment
$\vec{u} \perp \vec{N}_C$



Twist in the joint plane
about the joint line

The diagram is schematic. The theorem uses only the perpendicularity of the two rotation axes, not any particular polyhedron.

6 Parallel normals and straight-joint degeneracies

6.1 What the normals do and do not determine

If $\vec{N}_B = -\vec{N}_C$, then $\alpha_j = 0$ and the shortest alignment is the identity. The normals impose no condition on a spin about their common line. Once the out-vector reference fixes that spin,

$$\theta = \arccos(\cos \tau_{\text{eff}}) = |\tau_{\text{eff}}| \quad \text{for } -\pi \leq \tau_{\text{eff}} \leq \pi.$$

There is no failure of the formula at $\alpha_j = 0$. If instead $\vec{N}_B = \vec{N}_C$, then $\alpha_j = \pi$: choose any perpendicular half-turn axis. Every resulting face-opposing rotation has principal angle π, as the formula predicts. The normals alone do not select the entire transformation in either case.

6.2 A straight joint line hides subunit rotation

Let a screw motion rotate by any angle about an axis and translate along that axis. A point *on* the axis advances in a straight line, independently of the rotation. Thus $r = 0$ makes θ unidentifiable from the joint points. The original paper explicitly recognizes this in its straight-line special case: the point method leaves θ undefined, while orientation data can supply it. A listed value of zero for the straight-joint curve does not prove that the full rigid motion has $R = I$.

For the repository's face-center baseline, the full principal rotations are as follows. These are predictions of the actual rotation construction; they are not replacements for the catalog's convention for a degenerate joint curve.

Name	Face class	α_j	τ	Full R: θ	Catalog θ
Boxbeam	Cub O	0	-180	180	0
Staircase	Cub A	90	-180	180	0
Octabeam	Oct O	0	-60	60	0
Dodecabeam	Dod O	0	-108	108	0
Pearlshaft	Ico O	0	-60	60	0

All five have $r = 0$ in the catalog. The Staircase also shows that a straight-joint degeneracy can occur with nonparallel normals. Rotations of an unmarked solid may additionally be equivalent under its own symmetries; here R refers to the specified motion of a labeled subunit.

This corrects an interpretation in the exploratory discussion: the Dodecabeam and Pearlshaft rows do *not* establish cancelling phases of 108° and 60°. Their joint-point data cannot establish a phase at all. In the code baseline for these centered opposite faces, $R_0 = I$ and $\gamma_j = 0$; the imposed twist remains in R.

7 Scope, reproducibility, and subsequent work

The compact formula is

$$\boxed{\theta = 2\arccos\left(\cos\frac{\alpha_j}{2}\left|\cos\frac{\tau_{\text{eff}}}{2}\right|\right), \qquad \alpha_j = \arccos(-\vec{N}_B \cdot \vec{N}_C).}$$

Its origin is the classical composition of rotations about perpendicular axes. The contribution of this note is the application to the Joint Face Normal Method, an explicit account of its twist reference, and exact catalog expressions. No claim of priority for the rotation identity or of an exhaustive literature review is made. Related tetrahelix work is already discussed in the original paper [3].

For fixed τ_{eff}, neither the object length nor the translation vector enters θ. If a numerical twist reference is tied to joint positions, those positions can affect γ_j and must be included when translating that reference to τ_{eff}. Consequently a bare pair (α_j, τ) is sufficient only when the zero-twist convention has been specified.

The formula does not determine the screw axis, radius r, axial advance d, or helix angle ϕ. For those, the next step is to combine the rotation with $\vec{t}$, compute the screw axis, and locate a chosen joint relative to it. Nor does the formula ensure a physically nonintersecting stack or a closed torus. At $R = I$, the separate pure-translation or identity cases will also need to be retained.

Reproducible checks

The companion script `verify_joint_face_rotation.py` uses only Python's standard library. It checks the 23 exact expressions against the catalog's rounded values, enumerates the face-normal classes, compares matrix products with the half-angle law, and tests both the general reference phase and the five straight-joint cases. In the latter it verifies the full rotation and the collinearity of successive joints separately.

The source baseline is repository revision `8be3839de2204b966c8aedb7c28f6a6f5c18076e`, specifically `MathematicJournal/StackingHelix.tex` and `js/segment_helix_math.js`, including `adjoinPrism` and `PointAxis`. The original manuscript and implementation are not changed by this companion note. The standalone source compiles using ordinary pdfLaTeX and needs no external images or bibliography file.

Preparation. Developed through discussion between Robert L. Read and ChatGPT (OpenAI), with assistance in derivation, source inspection, verification, and drafting. This is a working note for colleague review.

References

[1] R. L. Read, *Calculating the Segmented Helix Formed by Repetitions of Identical Subunits thereby Generating a Zoo of Platonic Helices*, Mathematics **10**(14), 2533 (2022). doi:10.3390/math10142533.

[2] Public Invention, *segmented-helixes* source repository. GitHub repository. Revision and source paths are specified above.

[3] G. Sadler, F. Fang, R. Clawson, and K. Irwin, *Periodic Modification of the Boerdijk–Coxeter Helix (Tetrahelix)*, Mathematics **7**(10), 1001 (2019). doi:10.3390/math7101001.